Exercise Sheet (3) — Hydraulic Oils & Theoretical Background

Course: MEC 431 — Fluid Power Systems | **Chapter 2**
TA: Eng. Moh Zedan | **BUC** — Fall 2026/2027

Problem 1: Pressure & Power Losses in a Transmission Line (Laminar Flow)

Derivation:

Starting from the Darcy-Weisbach equation for pressure loss in a pipe:

> $\Delta P = \lambda \cdot \frac{L}{D} \cdot \frac{\rho v^2}{2}$ — Eq. (2.4)

For **laminar flow**, the friction coefficient is:

> $\lambda = \frac{64}{Re}$ where $Re = \frac{\rho v D}{\mu}$ — Eq. (2.5), (2.6)

The mean velocity in terms of flow rate:

> $v = \frac{Q}{A} = \frac{4Q}{\pi D^2}$ — Eq. (2.7)

Substituting (2.5), (2.6), (2.7) into (2.4):

> $\Delta P = \frac{64}{\frac{\rho v D}{\mu}} \cdot \frac{L}{D} \cdot \frac{\rho v^2}{2} = \frac{64\mu}{D^2} \cdot L \cdot \frac{v}{2} = \frac{32 \mu L v}{D^2}$

Replacing $v = \frac{4Q}{\pi D^2}$:

Pressure Loss (Laminar Flow):

$$\boxed{\Delta P = \frac{128 \mu L Q}{\pi D^4}} = R \cdot Q \quad \text{where} \quad R = \frac{128 \mu L}{\pi D^4} \text{ is the } \textbf{hydraulic resistance}$$

> **Analogy:** $\Delta P = R \cdot Q$ is analogous to Ohm's law $V = R \cdot I$. Both resistances dissipate energy.

Power Loss:

$$\boxed{\Delta N = \Delta P \cdot Q = R \cdot Q^2} \quad \text{[W]}$$

Problem 2: Pressure Increment Due to Temperature Increase in a Rigid Container

Derivation:

When a liquid is trapped in a rigid container and heated, two competing effects occur:

1. **Thermal expansion** tries to increase the volume: $\Delta V_T = V\alpha\Delta T$
2. **Compressibility** resists the expansion by building pressure: $\Delta V_P = -\frac{V\Delta P}{B}$

Since the container is **rigid**, the total volume change is zero:

$$\Delta V_{total} = \Delta V_T + \Delta V_P = 0$$

$$V\alpha\Delta T - \frac{V\Delta P}{B} = 0$$

Solving for ΔP:

$$\boxed{\Delta P = B \cdot \alpha \cdot \Delta T}$$

Where: B = Bulk modulus [Pa], α = Thermal expansion coefficient [K^{-1}], ΔT = Temperature increase [°C or K].

Example (from lecture): For $\alpha = 0.0007\ \mathrm{K}^{-1}, B = 1\ \mathrm{GPa}, \Delta T = 50^\circ\mathrm{C}$:
$\Delta P = 1 \times 10^9 \times 0.0007 \times 50 = 35\ \mathrm{MPa}$ — a very significant pressure rise!

Problem 3: Effect of Saturated Vapor Pressure on Hydraulic Systems

Cavitation occurs when the local pressure in the hydraulic system drops below the **saturated vapor pressure** of the oil at the operating temperature. This causes:

1. **Vapor bubble formation:** Dissolved gases and oil vapor form bubbles in low-pressure zones (e.g., pump suction, downstream of orifices).
2. **Bubble collapse (implosion):** When bubbles travel to high-pressure zones, they collapse violently, producing localized shock waves with pressures up to 1000 MPa.
3. **Erosion damage:** Repeated implosions erode metal surfaces of pumps, valves, and cylinders (pitting).
4. **Noise and vibration:** Cavitation produces characteristic rattling/knocking noise.
5. **Reduced efficiency:** Pump volumetric efficiency drops significantly because vapor pockets reduce effective flow.
6. **Oil degradation:** Extreme local temperatures during implosion degrade oil properties.

Prevention methods:

- Increase suction line diameter and decrease its length to reduce losses.
- Mount pump below the tank to increase static head at inlet.
- Use a booster pump or pressurized tank.

- Avoid inlet line filters and unnecessary restrictions.
- Maintain oil temperature within specified range (controls vapor pressure).

Problem 4: Percentage Volume Variation

Given: $B = 1.2\ \text{GPa} = 1.2 \times 10^9\ \text{Pa}, \quad \Delta P = 12\ \text{MPa} = 12 \times 10^6\ \text{Pa}$

Solution:

The bulk modulus is defined as:

$$B = \frac{\Delta P}{\Delta V / V} \quad \Rightarrow \quad \frac{\Delta V}{V} = \frac{\Delta P}{B}$$

$$\frac{\Delta V}{V} = \frac{12 \times 10^6}{1.2 \times 10^9} = 0.01$$

ΔV/V = 1% (volume decreases by 1%)

⚠️ **TA Note:** The negative sign indicates compression (volume decrease). Hydraulic oil is nearly incompressible — even 12 MPa (120 bar) only compresses it by 1%. This is why hydraulics can transmit power so rigidly.

Problem 5: Viscous Friction Coefficient of a Spool Valve

Given: $L_1 = L_2 = 12\ \text{mm} = 0.012\ \text{m},\ D = 10\ \text{mm} = 0.01\ \text{m},\ c = 3\ \mu\text{m} = 3 \times 10^{-6}\ \text{m},\ \mu = 0.03\ \text{N·s/m}^2$

Derivation:

When a spool moves axially with velocity $\dot{x}$, the oil in the radial clearance is sheared. Assuming linear velocity distribution in the small clearance:

Step 1: Shear stress on spool surface:
$\tau = \mu \cdot \frac{\dot{x}}{c}$

Step 2: Shear area (cylindrical surface of both lands):
$A_s = \pi D(L_1 + L_2)$

Step 3: Friction force:
$F_f = \tau \cdot A_s = \mu \cdot \frac{\dot{x}}{c} \cdot \pi D(L_1 + L_2) = f_v \cdot \dot{x}$

Therefore, the **viscous friction coefficient** is:

$$\boxed{f_v = \frac{\mu \pi D(L_1 + L_2)}{c}} \quad \text{[N·s/m]}$$

Calculation:

$$f_v = \frac{0.03 \times \pi \times 0.01 \times (0.012 + 0.012)}{3 \times 10^{-6}}$$

$$f_v = \frac{0.03 \times \pi \times 0.01 \times 0.024}{3 \times 10^{-6}} = \frac{0.03 \times 7.5398 \times 10^{-4}}{3 \times 10^{-6}} = \frac{2.262 \times 10^{-5}}{3 \times 10^{-6}}$$

f_v = 7.54 N·s/m

Problem 6: Pressure Upstream of Throttle Valve

Given: $Q = 15\ \text{L/min},\ \rho = 860\ \text{kg/m}^3,\ \mu = 0.0185\ \text{N·s/m}^2,\ A_t = 2\ \text{mm}^2,\ C_d = 0.611$

$D_1 = 15\ \text{mm},\ D_2 = 10\ \text{mm},\ D_3 = 12\ \text{mm},\ L_1 = 1.5\ \text{m},\ L_2 = 2.5\ \text{m},\ L_3 = 2\ \text{m}$

Step 1: Convert Units

$$Q = \frac{15}{60000} = 2.5 \times 10^{-4}\ \text{m}^3/\text{s}, \quad A_t = 2 \times 10^{-6}\ \text{m}^2$$

Step 2: Check Laminar Flow (Reynolds Number)

$$Re = \frac{4\rho Q}{\pi \mu D}$$

Line	D (mm)	Re	Flow Type
1	15	$\frac{4\times860\times2.5\times10^{-4}}{\pi\times0.0185\times0.015} = \frac{0.86}{8.64\times10^{-4}} = 995$	Laminar ✓
2	10	$\frac{0.86}{5.81\times10^{-4}} = 1480$	Laminar ✓
3	12	$\frac{0.86}{6.97\times10^{-4}} = 1233$	Laminar ✓

All Re < 2000 → Laminar flow assumption is valid ✓

Step 3: Pipe Pressure Losses (Hagen-Poiseuille)

$$\Delta P = \frac{128\mu L Q}{\pi D^4}$$

Line	D^4 (m⁴)	ΔP (Pa)	ΔP (kPa)
1	5.063×10^{-8}	$\frac{128\times0.0185\times1.5\times2.5\times10^{-4}}{\pi\times5.063\times10^{-8}} = 5,585$	**5.6**
2	1.0×10^{-8}	$\frac{128\times0.0185\times2.5\times2.5\times10^{-4}}{\pi\times1.0\times10^{-8}} = 47,109$	**47.1**
3	2.074×10^{-8}	$\frac{128\times0.0185\times2.0\times2.5\times10^{-4}}{\pi\times2.074\times10^{-8}} = 18,175$	**18.2**

Step 4: Throttle Valve Pressure Drop

From orifice equation: $Q = C_d A_t \sqrt{\frac{2\Delta P_t}{\rho}} \quad \Rightarrow \quad \Delta P_t = \frac{\rho Q^2}{2 C_d^2 A_t^2}$

$$\Delta P_t = \frac{860\times(2.5\times10^{-4})^2}{2\times(0.611)^2\times(2\times10^{-6})^2} = \frac{860\times6.25\times10^{-8}}{2\times0.3733\times4\times10^{-12}}$$

$$\Delta P_t = \frac{5.375\times10^{-5}}{2.987\times10^{-12}} = 17.99 \times 10^6\ \text{Pa}$$

$\Delta P_{\text{throttle}} \approx 18.0$ MPa

Step 5: Total Pressure Upstream of Throttle

Assuming circuit: Pump → Line 1 → Line 2 → Throttle → Line 3 → Tank

$$P_{upstream} = \Delta P_1 + \Delta P_2 + \Delta P_t + \Delta P_3$$

$$P_{upstream} = 5.6 + 47.1 + 18000 + 18.2 = 18,071 \text{ kPa}$$

P ≈ 18.07 MPa ≈ 180.7 bar

⚠️ **TA Note:** The pipe losses (≈ 71 kPa total) are negligible compared to the throttle drop (18 MPa). The small orifice area (2 mm²) creates the dominant restriction. This is the principle behind flow control valves.

Problem 7: Pump Inlet (Suction) Pressure

Given: $Q = 59\ \text{L/min},\ L = 1.5\ \text{m},\ D = 14\ \text{mm},\ \nu = 0.2\ \text{cm}^2/\text{s},\ \rho = 870\ \text{kg/m}^3$

Step 1: Convert Units

$$Q = \frac{59}{60000} = 9.833 \times 10^{-4}\ \text{m}^3/\text{s}, \quad \nu = 0.2 \times 10^{-4}\ \text{m}^2/\text{s} = 2 \times 10^{-5}\ \text{m}^2/\text{s}$$
$$\mu = \rho\nu = 870 \times 2 \times 10^{-5} = 0.0174\ \text{N·s/m}^2$$

Step 2: Calculate Velocity & Reynolds Number

$$v = \frac{Q}{A} = \frac{9.833\times10^{-4}}{\frac{\pi}{4}(0.014)^2} = \frac{9.833\times10^{-4}}{1.539\times10^{-4}} = 6.39\ \text{m/s}$$

$$Re = \frac{vD}{\nu} = \frac{6.39\times0.014}{2\times10^{-5}} = 4472$$

Re = 4472 > 2000 → Turbulent Flow! Cannot use Hagen-Poiseuille. Must use Darcy-Weisbach with turbulent friction factor.

Step 3: Friction Factor (Blasius Formula for Smooth Pipe)

$$\lambda = \frac{0.3164}{Re^{0.25}} = \frac{0.3164}{(4472)^{0.25}} = \frac{0.3164}{8.177} = 0.0387$$

Step 4: Pressure Loss (Darcy-Weisbach)

$$\Delta P = \lambda \cdot \frac{L}{D} \cdot \frac{\rho v^2}{2} = 0.0387 \times \frac{1.5}{0.014} \times \frac{870\times6.39^2}{2}$$

$$\Delta P = 0.0387 \times 107.14 \times \frac{870\times40.83}{2} = 4.146 \times 17,761 = 73,640\ \text{Pa}$$

Step 5: Pump Inlet Pressure

The tank is open (atmospheric), so the pump inlet pressure is:

$$P_{inlet} = P_{atm} - \Delta P_{friction} = 101,325 - 73,640 = 27,685\ \text{Pa}$$

P_{inlet} ≈ 27.7 kPa (abs) = 0.277 bar (abs)

⚠️ **TA Note:** This very low suction pressure (≈ 0.27 atm) is dangerously close to the oil's vapor pressure. This means **cavitation is likely!** Solutions: increase pipe diameter, shorten suction line, or mount pump below tank.

Problem 8: Pressure Losses in a Pipeline System

Given: $Q = 10\ \text{L/min},\ \rho = 850\ \text{kg/m}^3,\ L_1 = L_2 = 5\ \text{m},\ D_1 = 15\ \text{mm},\ D_2 = 10\ \text{mm},\ \nu = 21\ \text{cSt}$

Step 1: Convert Units

$$Q = \frac{10}{60000} = 1.667 \times 10^{-4}\ \text{m}^3/\text{s}, \quad \nu = 21 \times 10^{-6}\ \text{m}^2/\text{s},$$
$$\mu = \rho\nu = 850 \times 21 \times 10^{-6} = 0.01785\ \text{N·s/m}^2$$

Step 2: Check Reynolds Numbers

Pipe	D (m)	v (m/s)	Re = vD/ν	Flow
1	0.015	$\frac{1.667\times10^{-4}}{1.767\times10^{-4}} = 0.943$	$\frac{0.943\times0.015}{21\times10^{-6}} = 674$	Laminar ✓
2	0.010	$\frac{1.667\times10^{-4}}{7.854\times10^{-5}} = 2.122$	$\frac{2.122\times0.01}{21\times10^{-6}} = 1011$	Laminar ✓

Step 3: Pressure Losses (Hagen-Poiseuille)

Pipe 1: $\Delta P_1 = \frac{128\times0.01785\times5\times1.667\times10^{-4}}{\pi\times(0.015)^4} = \frac{1.904\times10^{-3}}{1.590\times10^{-7}} = 11,975\ \text{Pa}$

Pipe 2: $\Delta P_2 = \frac{128\times0.01785\times5\times1.667\times10^{-4}}{\pi\times(0.01)^4} = \frac{1.904\times10^{-3}}{3.142\times10^{-8}} = 60,590\ \text{Pa}$

Total Pressure Loss:

$$\Delta P_{total} = \Delta P_1 + \Delta P_2 = 11,975 + 60,590 = 72,565\ \text{Pa}$$

ΔP_{total} ≈ 72.6 kPa ≈ 0.726 bar

⚠️ **TA Note:** Pipe 2 (smaller diameter) causes **5× more loss** than Pipe 1, despite same length and flow. This is because $\Delta P \propto 1/D^4$. Reducing diameter from 15→10 mm increases resistance by $(15/10)^4 = 5.06$ times!

Problem 9: Piston Shift Due to Force Increase (Compressibility & Elasticity)

Given: $\Delta F = 105\text{ kN},\ A_p = 0.0178\text{ m}^2,\ h = 1\text{ mm (pipe wall)},\ d = 15\text{ mm (pipe ID)}$

$L = 6\text{ m (pipe)},\ t = 5\text{ mm (cyl. wall)},\ B = 1.2\text{ GPa},\ E = 210\text{ GPa},\ x_0 = 0.6\text{ m}$

Step 1: Pressure Increase

$$\Delta P = \frac{\Delta F}{A_p} = \frac{105\times10^3}{0.0178} = 5.899\times10^6\text{ Pa} \approx 5.9\text{ MPa}$$

Step 2: Cylinder Bore Diameter

$$A_p = \frac{\pi}{4}D_{cyl}^2 \quad\Rightarrow\quad D_{cyl} = \sqrt{\frac{4\times0.0178}{\pi}} = 0.1506\text{ m} = 150.6\text{ mm}$$

Step 3: Fluid Volumes

Cylinder volume: $V_{cyl} = A_p \times x_0 = 0.0178 \times 0.6 = 0.01068\text{ m}^3$

Pipe volume: $V_{pipe} = \frac{\pi}{4}d^2L = \frac{\pi}{4}(0.015)^2 \times 6 = 1.06\times10^{-3}\text{ m}^3$

Step 4: Effective Bulk Modulus (accounting for wall elasticity)

For a thin-walled container under internal pressure:

$$\frac{1}{B_{eff}} = \frac{1}{B} + \frac{D_{container}}{2\cdot t_{wall}\cdot E}$$

For the pipe:

$$\frac{1}{B_{eff,pipe}} = \frac{1}{1.2\times10^9} + \frac{0.015}{2\times0.001\times210\times10^9} = 8.333\times10^{-10} + 3.571\times10^{-11} = 8.690\times10^{-10}$$

$$B_{eff,pipe} = 1.151\times10^9\text{ Pa}$$

For the cylinder:

$$\frac{1}{B_{eff,cyl}} = \frac{1}{1.2\times10^9} + \frac{0.1506}{2\times0.005\times210\times10^9} = 8.333\times10^{-10} + 7.171\times10^{-11} = 9.050\times10^{-10}$$

$$B_{eff,cyl} = 1.105\times10^9\text{ Pa}$$

Step 5: Volume Changes

$$\Delta V_{pipe} = \frac{V_{pipe}\cdot\Delta P}{B_{eff,pipe}} = \frac{1.06\times10^{-3}\times5.9\times10^6}{1.151\times10^9} = 5.43\times10^{-6}\text{ m}^3$$

$$\Delta V_{cyl} = \frac{V_{cyl}\cdot\Delta P}{B_{eff,cyl}} = \frac{0.01068\times5.9\times10^6}{1.105\times10^9} = 5.70\times10^{-5}\text{ m}^3$$

Step 6: Piston Shift

$$\Delta V_{total} = \Delta V_{pipe} + \Delta V_{cyl} = 5.43 \times 10^{-6} + 5.70 \times 10^{-5} = 6.25 \times 10^{-5}\ \text{m}^3$$

$$\Delta x = \frac{\Delta V_{total}}{A_p} = \frac{6.25 \times 10^{-5}}{0.0178} = 3.51 \times 10^{-3}\ \text{m}$$

Δx ≈ 3.51 mm

⚠️ **TA Note:** This 3.5 mm "lost motion" is entirely due to oil compression and elastic expansion of the pipe/cylinder walls. The cylinder contributes ~91% of the total shift (larger volume), while the pipe contributes only ~9%. This is why high-precision hydraulic systems use thick-walled cylinders and short, rigid pipes.